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Is blood the new stool? Status of blood tests for CRC screening

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ABSTRACT

Colorectal cancer (CRC) screening has proven to decrease CRC incidence and mortality, with many countries providing faecal immunochemical tests (FIT) in screening programs. Effectiveness, however, is limited in the setting of low participation rates, which has prompted growing research into blood-based screening. While blood testing is anticipated to improve participation due to greater population acceptability, its readiness for widespread implementation remains unclear, and comparative accuracy data against FIT is needed. This article provides a systematic review and meta-analysis on blood-based CRC screening tests that have been directly compared to FIT, and summarises the evidence for broader program considerations, including analytical quality, consumer preferences, participation rates and cost effectiveness. The findings show that blood tests currently lack the sensitivity and cost-effectiveness to replace FIT in screening programs, and that their use should be limited to individuals unable or unwilling to complete FIT.

1. Introduction

Colorectal cancer (CRC) is the third most diagnosed cancer and the second leading cause of cancer-related deaths worldwide [1]. CRC typically develops slowly through the adenoma-carcinoma or serrated pathways [2], providing opportunity for early detection and intervention with screening. One approach involves a two-step process-measuring faecal haemoglobin (Hb) levels and referring individuals exceeding a set threshold to colonoscopy. Initial CRC screening programs used guaiac-based faecal occult blood tests (gFOBT) which were shown to reduce both CRC incidence and mortality [3]. However, due to the impact of dietary factors and medication on gFOBT accuracy [4], many countries have now replaced gFOBT and implemented faecal immunochemical tests (FIT) [5]. FIT is simpler to perform and has a superior specificity and sensitivity (for both CRC and precancerous lesions) [6]. Its use has led to a decrease in CRC incidence (advanced and overall) [7,8] and a 62 % decrease in mortality [9].

Despite the improved participation with FIT over gFOBT [10], many countries still report suboptimal uptake of FIT [5], falling short of the 80 % uptake target [11]. Barriers such as aversion to faecal sampling [12], have prompted the development and introduction of alternative non-invasive screening tests. Blood tests are an obvious test due to high public preference for blood tests over stool-based sampling for CRC screening [13]. Among blood-based tests, the most studied is Epi proColon, which detects methylation of *SEPTIN 9* (*SEPT9*) in circulating tumour DNA (ctDNA) [14]. This test was approved by the U.S. Food and Drug Administration (FDA) in April 2016 for average risk individuals who decline other CRC screening options. However, it is not recommended as a first-line screening test. More recently, in July 2024, the FDA approved the blood test Shield (Guardant Health) as a primary CRC screening tool. Shield analyses cell free DNA (cfDNA) for aberrant methylation, genomic alterations, and fragmentomic signatures [15].

The growth in the number of blood-based tests to detect CRC raises a critical question: is blood the new stool? Do blood-based tests have the diagnostic potential to improve CRC screening, by providing an accurate test with higher uptake than FIT? To answer this question, it is important to understand what characteristics are required of a CRC screening test. Such characteristics will vary between screening jurisdictions and often be dictated by local public health policy [16]. Accuracy of new diagnostic tests do not necessarily need to be assessed in a randomised controlled trial, providing that the new test is safe, and has higher specificity and a similar sensitivity to the old test [17]. A recent consensus piece on requirements for diagnostic tests highlight that any new non-invasive biomarker tests should be compared to FIT [18], which is widely regarded as the gold standard among two-step screening tests.

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Test accuracy is not the only consideration. Screening logistics, population preferences, participation (acceptance and adherence) rates, and cost-effectiveness are also crucial to consider (Table 1).

It is not clear how many blood-based tests have undergone such a comprehensive evaluation, or at a minimum, been compared against validated FIT assays. Therefore, in this article we present a systematic review and meta-analysis analysis of blood tests that have had diagnostic accuracy evaluated in head-to-head comparisons against FIT. As accuracy data alone do not determine clinical utility [22], we have also reviewed the literature on the broader context to assess whether blood-based CRC screening is ready for widespread adoption.

2. Diagnostic accuracy of blood-based tests compared to FIT

We conducted a systematic review and meta-analysis following the Cochrane guidelines for Systematic Reviews for diagnostic accuracy and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.

2.1. Methodology

2.1.1. Search and screen strategy

MEDLINE, Web of Science and Scopus were searched on April 3rd, 2025, using keywords and Medical Subject Headings outlined in Supplementary Table 1. Studies were eligible if they evaluated the performance (sensitivity and specificity) of both FIT and a blood biomarker test for colorectal neoplasia (including CRC, advanced adenomas, and/or advanced serrated lesions). The FIT and blood test must have been assessed in individuals ≥18 years that underwent colonoscopy (within 1 year post-test), with sufficient data to extract or calculate both sensitivity and specificity. Studies that only included participants with a positive FIT as the indication for colonoscopy or non-immunochemical faecal tests (e.g. guaiac or chemical based), were excluded. Eligible studies included randomised controlled trials, case-control, prospective and retrospective studies. Reviews, case reports, editorials, conference abstracts, study protocols, animal models and *in vitro* studies were excluded. There was no year limit for publication of the study. Studies that were not published in English were excluded as there were no resources for language translation.

The screening of the title, abstract and full text for eligibility were conducted independently and in a blinded manner by ELS and JMW. Discrepancies were then resolved by discussion.

Table 1
Considerations in evaluating new colorectal cancer screening tests.

Attributes to consider	Comments
Accuracy- sensitivity and specificity	The US Centers for Medicare & Medicaid Services (CMS) will cover the costs of blood-based screening tests providing that they demonstrate a sensitivity of ≥74 % for colorectal cancer and specificity of ≥90 % [19]. However, no sensitivity threshold for detecting precancerous lesions (e.g. adenomas) have been specified.
Participation	Screening participation is influenced by population preferences and perceived barriers. The National Colorectal Cancer Roundtable set a goal of 80 % participation by 2018. This was projected to save over 203,000 lives in the US by 2030 [20].
Long-term adherence	Re-participation is also an essential consideration as a single round of colorectal cancer screening will not provide lifelong protection. Evidence shows that risk reduction improves with successive rounds of screening with faecal immunochemical tests [21].
Logistics of program implementation	Program delivery considerations include the frequency of screening test (e.g. annually, biennially), and protocols following not only a positive screening test but also guidance on follow-up if the colonoscopy is negative (e.g. the need for further diagnostic evaluation, or guidance on when to return to screening).
Cost effectiveness	Modelling can inform cost effectiveness of new colorectal cancer screening tests, but it must be informed by test accuracy, participation rates and positivity rates. These factors determine workload and overall screening effectiveness.
Program effectiveness	Screening programs should be assessed over repeated screening rounds for impact on colorectal cancer incidence and mortality, as well as interval cancer findings.

2.1.2. Data extraction

ELS and JMW extracted accuracy data from included studies, including sensitivity, specificity, true positive (TP), true negative (TN), false positive (FP), and false negative (FN). Demographic and clinical information collected included country, age, sex, total number of participants, number of participants included in the accuracy measure, study design, colonoscopy indication, time from test to colonoscopy, and definition of neoplasia. Data collected on the FIT included brand, qualitative/quantitative status, and faecal Hb positivity threshold. Data collected on the blood-based tests included biomarker type, assay technique, whether it was serum or plasma, and positivity threshold.

2.1.3. Quality assessment and risk of bias

ELS and JMW independently assessed risk of bias (RoB) and applicability of each individual test (FIT and blood) as well as the RoB in the comparison of FIT versus blood-based tests using Quality Assessment of Diagnostic Accuracy Studies, 2nd edition (QUADAS-2) tool [23]. Discrepancies were resolved by discussion.

2.1.4. Data analysis

Meta-analysis was performed in R using the packages metafor (bivariate or multivariable random effects) and mada (paired-difference bivariate reitsma model). Where analyses included multiple data points from the same outcome (i.e. multiple thresholds or biomarkers), multivariable analyses were used. Multivariable and bivariate random effects meta-analysis was performed on the sensitivity, specificity and positive predictive value (PPV) using TP, TN, FP, and FN data as the input. Data are presented as sensitivity, specificity, PPV and pooled sensitivity, pooled specificity and pooled PPV with 95 % confidence intervals (CI). Where variance for a study was zero, a continuity correction of 0.5 was applied and the meta-analysis repeated. Heterogeneity was tested using the restricted maximum-likelihood (REML) estimator for τ^2 and I^2. The *Q*-Profile method was used for the CI of τ^2 and τ, with calculation of I^2 based on *Q*.

Meta-analysis was performed for FIT and blood accuracy (sensitivity and specificity) for each outcome of CRC, precancerous lesions (focussing on advanced adenoma and/or high risk sessile serrated lesion), and early and late-stage CRC. A sub-analysis was conducted for sensitivity and specificity of methylated *SEPT9* for CRC. A sub-analysis of FIT and blood test PPV for CRC was conducted in prospectively designed studies only, since case-control studies do not accurately reflect the true prevalence of neoplasia in a population.

A pairwise meta-analysis of the head-to-head comparison between FIT and blood tests for detection of CRC (all stages) and precancerous lesions was performed and plotted using summary receiver operating characteristic curve (SROC). Wilcoxon signed rank test with continuity correction was used to compare sensitivity and specificity in a pairwise fashion between FIT and blood-based tests. *P*-values <0.05 were considered statistically significant.

2.2. Results

2.2.1. Included studies

A total of 11,400 studies were identified for title and abstract screening. After importing into EndNote, 2574 duplicates were identified, with 8826 remaining studies imported into Covidence (Fig. 1).

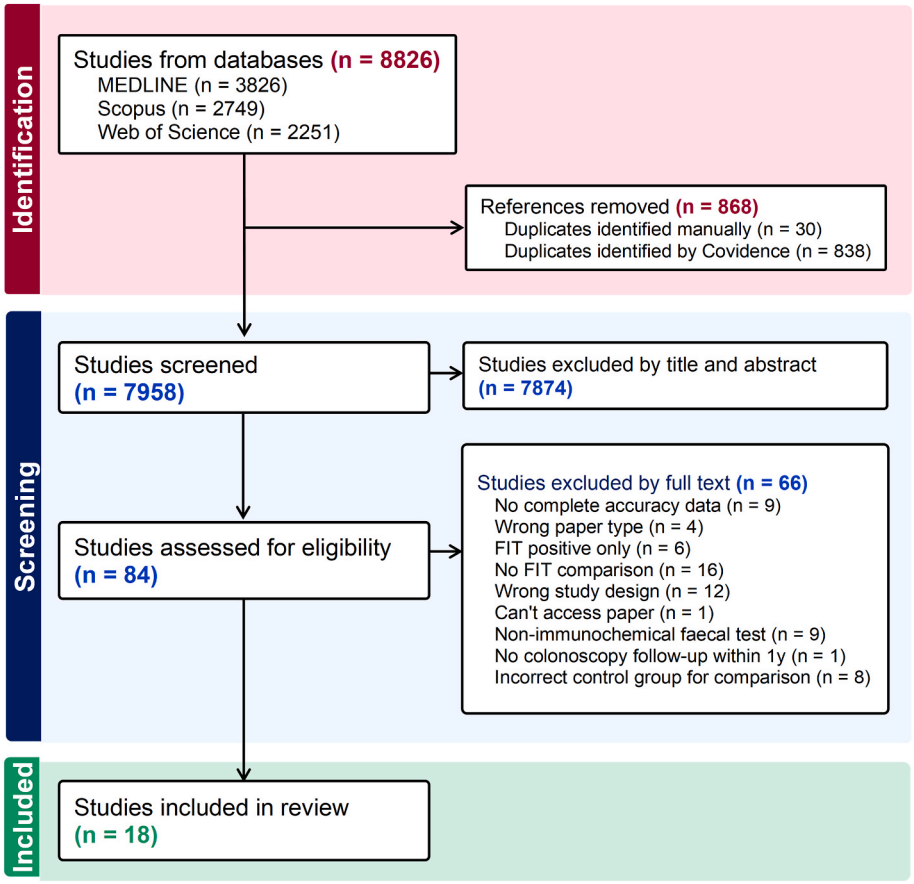


Fig. 1. PRISMA flow chart for study selection, screening and final included studies for the systematic review and meta-analysis.

After exclusions, 84 studies were reviewed for full text screening (Supplementary Table 2), with 18 studies eligible for inclusion in the systematic review and meta-analysis (Fig. 1).

2.2.2. Study characteristics

A summary of the demographics, study design, and test details is presented in Table 2 and Supplementary Table 3.

Eight (44.4 %) studies were conducted in China, with the remaining studies conducted in Japan (n = 1), Taiwan (n = 1), Australia (n = 2), Egypt (n = 1), Germany (n = 2), Spain (n = 2), and the United States (n = 1), with n = 10 (55.6 %) having a case-control design and n = 8 (44.4 %) being prospective cohort studies. Four of the studies reported including a cohort undergoing screening colonoscopy, but the majority of studies included cases with diagnostic or surveillance colonoscopies (Supplementary Table 3). A total of 8380 participants were assessed for accuracy of FIT and 7987 assessed for accuracy of blood biomarkers. CRC was the diagnostic outcome in 15/18 (83.3 %) of studies and pre-cancerous lesions were measured in 14/18 studies. Of the studies assessing accuracy for CRC, 5/15 (33.3 %) assessed the accuracy for early-stage CRC (stage 0, I, and/or II) and 6/15 (40 %) for late-stage disease (stage III and IV). One study [26] assessed the accuracy for sessile serrated lesions separately. Specificity assessments varied between studies, with some calculated in individuals with no evidence of any pathology, with other calculations based on absence of CRC.

Types of FITs included seven studies with qualitative tests (OnSite™ FOB-Hi Rapid Test, Gold gel stripe, Colloidal gold test, InSure), ten studies with quantitative tests (OC-Sensor, One-Step Fecal Occult Blood Diagnostic Kit, OC-HEMODIA, RIDASCREEN, ADVIA Centaur® Analyzer, FIT FOB Gold) and one study that did not report type [41]. All but four studies reported a positivity f-Hb threshold [29,30,35,41]. The most common type of blood biomarker assessed was circulating tumour DNA (ctDNA) methylation assays (n = 10 studies, 55.6 %) measuring *SEPT9* [25,27,28,30,35,40], *BCAT1* and/or *IKZF1* [26,36], *TWIST1* [34], and a multi-gene next generation sequencing (NGS) panel [41]. Protein biomarkers were assessed in five different studies and included TAP [29], NDKA [32], sCD26 [33,37],CRP and TIMP-1 [37], CEA and CA19-9 [40] and a 5-biomarker panel measuring CEA, ferritin, seprase, osteopontin, and anti-p53 antibody [39]. Circulating tumour cells (CTCs) [38], miRNA [31] and mRNA [24] were assessed in one study for each biomarker. Plasma was used to assess ctDNA methylation in 8 studies with two studies using serum. Protein, miRNA, and mRNA were all assessed in serum, and CTCs were assessed in whole blood.

2.2.3. Study quality assessment

RoB and QUADAS-2 assessment are summarised in Supplementary Fig. 1. For domain 1 patient selection (Supplementary Fig. 1a) most studies (16/18; 88.9 %) scored well for applicability of both the FIT and blood test, but half (9/18) of all studies for both FIT and blood test scored high for RoB, leading to only 7/18 studies (38.9 %) having a low RoB in the head-to-head comparison. The index test (domain 2; Supplementary Fig. 1b) had a low risk of concern for applicability for FIT and blood test (77.8 % and 88.9 %, respectively), low risk of bias in most FIT studies (12/18 66.7), but high risk of bias in 10/18 (55.6 %) blood-based test studies, leading to an overall 13/18 (72.2 %) of studies having a high risk of bias in the head-to head comparison. There was low risk of concern for applicability and RoB for domain 3 the reference standard (Supplementary Fig. 1c) for FIT, blood and the comparison (>77.8 % of studies), but most studies had a high RoB for domain 4 flow and timing for FIT (61.1 %), blood test (72.2 %) and the head-to-head comparison (77.8 %; (Supplementary Fig. 1d). All studies in the review demonstrated at least one or more RoB or applicability concerns across all four domains.

Table 2
Characteristics and test sensitivity (Sn) and specificity (Sp) from the 18 included studies.

			Faecal immunochemical test					Blood-based test						
Author	Study design	Disease state	Brand	Positivity threshold	N	Sn (%)	Sp (%)	Blood	Biomarker	Biomarker class	Positivity threshold	N	Sn (%)	Sp (%)
Abdel Ghafar [24]	Case-Control	CRC	OnSite™ FOB-Hi Rapid Test (CTK Biotech)	3.5 μg/g	164	50.0	42.3	Serum	Metadherin	mRNA	4	164	91.9	92.3
	Case-Control	CRC	OnSite™ FOB-Hi Rapid Test (CTK Biotech)	3.5 μg/g	164	50.0	42.3	Serum	CEA	protein	8 ng/mL	164	60.5	79.4
	Case-Control	CRC	OnSite™ FOB-Hi Rapid Test (CTK Biotech)	3.5 μg/g	164	50.0	42.3	Serum	CA 19.9	protein	29 U/mL	164	58.1	79.5
Chen [25]	Case-Control	CRC	not reported	0.2 μg/ml[a]	60	84.3	55.6	plasma	*SEPT9*	ctDNA methylation	1/3 replicates	60	47.1	88.9
	Case-Control	Stage 0, I, and II	not reported	0.2 μg/ml[a]	32	69.6	55.6	plasma	*SEPT9*	ctDNA methylation	1/3 replicates	32	34.8	88.9
	Case-Control	stage III and IV	not reported	0.2 μg/ml[a]	37	96.4	55.6	plasma	*SEPT9*	ctDNA methylation	1/3 replicates	37	57.1	88.9
Cock [26]	Prospective	Advanced adenoma	OC-Sensor (Eiken Chemical Company)	20 μg/g	1882	28.7	91.6	plasma	*BCAT1/IKZF1*	ctDNA methylation	presence of either biomarker	1403	8.9	93.0
	Prospective	Advanced sessile serrated lesion	OC-Sensor (Eiken Chemical Company)	20 μg/g	1882	17.4	91.6	plasma	*BCAT1/IKZF1*	ctDNA methylation	presence of either biomarker	1403	11.8	93.0
	Prospective	Advanced adenoma	OC-Sensor (Eiken Chemical Company)	10 μg/g	1882	43.4	87.4	plasma	*BCAT1/IKZF1*	ctDNA methylation	presence of either biomarker	1403	8.9	93.0
	Prospective	Advanced sessile serrated lesion	OC-Sensor (Eiken Chemical Company)	10 μg/g	1882	21.7	87.4	plasma	*BCAT1/IKZF1*	ctDNA methylation	presence of either biomarker	1403	11.8	93.0
Jin [27]	Prospective	CRC	gold gel stripe (WanhuaPuman, Inc. Beijing, China)	0.2 μg/mL[a]	177	58.0	96.3	plasma	*SEPT9*	ctDNA methylation	2/3 rule	177	76.8	96.3
	Prospective	Advanced adenoma	gold gel stripe (WanhuaPuman, Inc. Beijing, China)	0.2 μg/mL[a]	177	28.6	96.3	plasma	*SEPT9*	ctDNA methylation	2/3 rule	177	25.7	96.3
Johnson [28]	Prospective	CRC	OC-Sensor (Eiken Chemical Company)	20 μg/g	290	68.0	97.4	plasma	*SEPT9*	ctDNA methylation	1/3 replicates	290	72.2	80.8
Liu [29]	Retrospective case control	CRC	colloidal gold test	not reported	99	89.6	73.3	whole blood	TAP	protein	agglutination particle area≥ 121 μm2	99	100.0	86.7
	Retrospective case control	Advanced adenoma	colloidal gold test	not reported	77	53.2	73.3	whole blood	TAP	protein	agglutination particle area≥ 121 μm2	77	80.9	86.7
	Retrospective case control	Stage I	colloidal gold test	not reported	40	90.0	73.3	whole blood	TAP	protein	agglutination particle area≥ 121 μm2	40	100.0	86.7
	Retrospective case control	Stage II	colloidal gold test	not reported	54	83.3	73.3	whole blood	TAP	protein	agglutination particle area≥ 121 μm2	54	100.0	86.7
	Retrospective case control	Stage III	colloidal gold test	not reported	55	92.0	73.3	whole blood	TAP	protein	agglutination particle area≥ 121 μm2	55	100.0	86.7

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Table 2 (*continued*)

Author	Study design	Disease state	Faecal immunochemical test					Blood-based test						
			Brand	Positivity threshold	N	Sn (%)	Sp (%)	Blood	Biomarker	Biomarker class	Positivity threshold	N	Sn (%)	Sp (%)
	Retrospective case control	Stage IV	colloidal gold test	not reported	40	100.0	73.3	whole blood	TAP	protein	agglutination particle area≥ 121 μm2	40	100.0	86.7
Lu [30]	Case-Control	CRC	InSure FIT (BioChain)	not reported	326	87.7	79.6	plasma	*SEPT9*	ctDNA methylation	Ct value ≤ 41	326	76.5	87.8
Luo [31]	Case-Control	CRC	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	182	69.3	96.6	serum	miR-18a	miRNA	Set by model with specificity at 96.4 %	182	2.7	96.6
	Case-Control	Advanced adenoma	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	125	26.0	97.2	serum	miR-18a	miRNA	Set by model with specificity at 96.4 %	125	11.8	97.2
	Case-Control	CRC	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	182	69.3	96.6	serum	miR-20a	miRNA	Set by model with specificity at 96.4 %	182	5.1	96.6
	Case-Control	Advanced adenoma	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	125	26.0	97.2	serum	miR-20a	miRNA	Set by model with specificity at 96.4 %	125	13.2	97.2
	Case-Control	CRC	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	182	69.3	96.6	serum	miR-21	miRNA	Set by model with specificity at 96.4 %	182	8.8	96.6
	Case-Control	Advanced adenoma	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	125	26.0	97.2	serum	miR-21	miRNA	Set by model with specificity at 96.4 %	125	7.2	97.2
	Case-Control	CRC	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	182	69.3	96.6	serum	miR-29a	miRNA	Set by model with specificity at 96.4 %	182	0.3	96.6
	Case-Control	Advanced adenoma	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	125	26.0	97.2	serum	miR-29a	miRNA	Set by model with specificity at 96.4 %	125	5.1	97.2
	Case-Control	CRC	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	182	69.3	96.6	serum	miR-92a	miRNA	Set by model with specificity at 96.4 %	182	7.1	96.6
	Case-Control	Advanced adenoma	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	125	26.0	97.2	serum	miR-92a	miRNA	Set by model with specificity at 96.4 %	125	17.2	97.2
	Case-Control	CRC	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	182	69.3	96.6	serum	miR-106b	miRNA	Set by model with specificity at 96.4 %	182	7.5	96.6
	Case-Control	Advanced adenoma	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	125	26.0	97.2	serum	miR-106b	miRNA	Set by model with specificity at 96.4 %	125	11.7	97.2
	Case-Control	CRC	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	182	69.3	96.6	serum	miR-133a	miRNA	Set by model with specificity at 96.4 %	182	5.1	96.6
	Case-Control	Advanced adenoma	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	125	26.0	97.2	serum	miR-133a	miRNA	Set by model with specificity at 96.4 %	125	4.9	97.2
	Case-Control	CRC	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	182	69.3	96.6	serum	miR-143	miRNA	Set by model with specificity at 96.4 %	182	11.5	96.6

(*continued on next page*)

Table 2 (*continued*)

			Faecal immunochemical test					Blood-based test						
Author	Study design	Disease state	Brand	Positivity threshold	N	Sn (%)	Sp (%)	Blood	Biomarker	Biomarker class	Positivity threshold	N	Sn (%)	Sp (%)
	Case-Control	Advanced adenoma	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	125	26.0	97.2	serum	miR-143	miRNA	Set by model with specificity at 96.4 %	125	11.3	97.2
	Case-Control	CRC	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	182	69.3	96.6	serum	miR-145	miRNA	Set by model with specificity at 96.4 %	182	9.7	96.6
	Case-Control	Advanced adenoma	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	125	26.0	97.2	serum	miR-145	miRNA	Set by model with specificity at 96.4 %	125	11.6	97.2
	Case-Control	CRC	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	182	69.3	96.6	serum	miR-181b	miRNA	Set by model with specificity at 96.4 %	182	5.3	96.6
	Case-Control	Advanced adenoma	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	125	26.0	97.2	serum	miR-181b	miRNA	Set by model with specificity at 96.4 %	125	11.7	97.2
	Case-Control	CRC	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	182	69.3	96.6	serum	miR-342-3p	miRNA	Set by model with specificity at 96.4 %	182	0.6	96.6
	Case-Control	Advanced adenoma	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	125	26.0	97.2	serum	miR-342-3p	miRNA	Set by model with specificity at 96.4 %	125	6.1	97.2
	Case-Control	CRC	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	182	69.3	96.6	serum	miR-532-3p	miRNA	Set by model with specificity at 96.4 %	182	3.2	96.6
	Case-Control	Advanced adenoma	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	125	26.0	97.2	serum	miR-532-3p	miRNA	Set by model with specificity at 96.4 %	125	5.3	97.2
	Case-Control	CRC	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	182	69.3	96.6	serum	miR-21, 92a, 106b, and 143	miRNA	Set by model with specificity at 96.4 %	182	26.7	88.3
	Case-Control	Advanced adenoma	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	125	26.0	97.2	serum	miR-21, 92a, 106b, and 143	miRNA	Set by model with specificity at 96.4 %	125	17.6	69.1
	Case-Control	CRC	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	182	69.3	96.6	serum	miR-21, 92a, 106b, 143, and 181b	miRNA	Set by model with specificity at 96.4 %	182	26.9	85.9
	Case-Control	Advanced adenoma	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	125	26.0	97.2	serum	miR-21, 92a, 106b, 143, and 181b	miRNA	Set by model with specificity at 96.4 %	125	11.3	86.1
	Case-Control	CRC	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	182	65.1	97.8	serum	multi-panel	miRNA	Set by model with specificity at 96.4 %	182	11.9	97.8
	Case-Control	Advanced adenoma	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	125	30.1	97.8	serum	multi-panel	miRNA	Set by model with specificity at 96.4 %	125	16.9	97.8
	Case-Control	stage I and II	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	157	58.1	97.8	serum	multi-panel	miRNA	Set by model with specificity at 96.4 %	157	20.9	97.8
	Case-Control	stage III and IV	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	149	75.1	97.8	serum	multi-panel	miRNA	Set by model with specificity at 96.4 %	149	20.5	97.8

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Table 2 (*continued*)

			Faecal immunochemical test					Blood-based test						
Author	Study design	Disease state	Brand	Positivity threshold	N	Sn (%)	Sp (%)	Blood	Biomarker	Biomarker class	Positivity threshold	N	Sn (%)	Sp (%)
	Case-Control	CRC	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	182	65.1	97.8	serum	miR-21, 92a, 106b, 143, and 181b	miRNA	Set by model with specificity at 96.4 %	182	16.8	97.8
	Case-Control	Advanced adenoma	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	125	30.1	97.8	serum	miR-21, 92a, 106b, 143, and 181b	miRNA	Set by model with specificity at 96.4 %	125	17.6	97.8
	Case-Control	stage I and II	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	157	58.1	97.8	serum	miR-21, 92a, 106b, 143, and 181b	miRNA	Set by model with specificity at 96.4 %	157	19.6	97.8
	Case-Control	stage III and IV	One-Step Fecal Occult Blood Diagnostic Kit, Suzhou Hailu Biotech Co., Ltd, China)	0.2 μg/mL[a]	149	75.1	97.8	serum	miR-21, 92a, 106b, 143, and 181b	miRNA	Set by model with specificity at 96.4 %	149	15.1	97.8
Otero-Estévez [32]	Prospective	Advanced neoplasia	OC-Sensor (Eiken Chemical Company)	20 μg/g	511	36.8	98.2	Serum	NDKA	protein	≥62.59 pg/mL	509	63.2	65.2
	Prospective	Advanced neoplasia	OC-Sensor (Eiken Chemical Company)	20 μg/g	511	36.8	98.2	Serum	NDKA	protein	≥76.07 pg/mL	509	29.8	85.0
	Prospective	Advanced neoplasia	OC-Sensor (Eiken Chemical Company)	20 μg/g	511	36.8	98.2	Serum	NDKA	protein	≥82.21 pg/mL	509	24.6	90.1
	Prospective	Advanced neoplasia	OC-Sensor (Eiken Chemical Company)	20 μg/g	511	36.8	98.2	Serum	NDKA	protein	≥94.79 pg/mL	509	17.5	95.2
	Prospective	Advanced adenoma	OC-Sensor (Eiken Chemical Company)	20 μg/g	507	32.1	98.2	Serum	NDKA	protein	≥62.59 pg/mL	509	62.3	65.2
	Prospective	Advanced adenoma	OC-Sensor (Eiken Chemical Company)	20 μg/g	507	32.1	98.2	Serum	NDKA	protein	≥76.07 pg/mL	509	30.2	85.0
	Prospective	Advanced adenoma	OC-Sensor (Eiken Chemical Company)	20 μg/g	507	32.1	98.2	Serum	NDKA	protein	≥82.21 pg/mL	509	24.5	90.1
	Prospective	Advanced adenoma	OC-Sensor (Eiken Chemical Company)	20 μg/g	507	32.1	98.2	Serum	NDKA	protein	≥94.79 pg/mL	509	17.0	95.2
Otero-Estévez [33]	Prospective	Advanced adenoma	OC-Sensor (Eiken Chemical Company)	20 μg/g	512	32.1	98.0	Serum	sCD26	protein	≤330 ng/mL	512	39.6	90.2
	Prospective	Advanced neoplasia	OC-Sensor (Eiken Chemical Company)	20 μg/g	516	36.8	98.3	Serum	sCD26	protein	≤280 ng/mL	516	42.1	90.2
	Prospective	Advanced adenoma	OC-Sensor (Eiken Chemical Company)	20 μg/g	512	32.1	98.0	Serum	sCD26	protein	≤330 ng/mL	512	28.3	95.2
	Prospective	Advanced neoplasia	OC-Sensor (Eiken Chemical Company)	20 μg/g	516	36.8	98.3	Serum	sCD26	protein	≤280 ng/mL	516	28.1	95.2
Suehiro [34]	Prospective	Advanced adenoma	OC-HEMODIA/OC-Sensor (Eiken Chemical Company)	20ug/g	95	24.3	88.0	serum	TWIST1	ctDNA methylation	2.8 copies	95	30.0	92.0
	Prospective	Stage I	OC-HEMODIA/OC-Sensor (Eiken Chemical Company)	20ug/g	39	28.6	88.0	serum	TWIST1	ctDNA methylation	2.8 copies	39	50.0	92.0
	Prospective	CRC	OC-HEMODIA/OC-Sensor (Eiken Chemical Company)	20ug/g	43	44.4	88.0	serum	TWIST1	ctDNA methylation	2.8 copies	43	44.4	92.0
Sun [35]	Case-Control	CRC	colloidal gold-antibody	not reported	557	58.7	91.9	plasma	*SEPT9*	ctDNA methylation	Ct value ≤ 41	557	73.0	94.5
Symonds [36]	Prospective	CRC	OC-Sensor (Eiken Chemical Company)	10 μg/g	933	78.8	86.3	plasma	BCAT1/IKZF1	ctDNA methylation	1 replicate positive for any biomarker	933	62.1	92.3

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Table 2 (*continued*)

			Faecal immunochemical test					Blood-based test						
Author	Study design	Disease state	Brand	Positivity threshold	N	Sn (%)	Sp (%)	Blood	Biomarker	Biomarker class	Positivity threshold	N	Sn (%)	Sp (%)
	Prospective	Stage I	OC-Sensor (Eiken Chemical Company)	10 μg/g	884	76.5	86.3	plasma	BCAT1/IKZF1	ctDNA methylation	1 replicate positive for any biomarker	884	41.2	92.3
	Prospective	Stage II	OC-Sensor (Eiken Chemical Company)	10 μg/g	892	80.0	86.3	plasma	BCAT1/IKZF1	ctDNA methylation	1 replicate positive for any biomarker	892	76.0	92.3
	Prospective	Stage III	OC-Sensor (Eiken Chemical Company)	10 μg/g	884	76.0	86.3	plasma	BCAT1/IKZF1	ctDNA methylation	1 replicate positive for any biomarker	884	58.8	92.3
	Prospective	Stage IV	OC-Sensor (Eiken Chemical Company)	10 μg/g	874	85.7	86.3	plasma	BCAT1/IKZF1	ctDNA methylation	1 replicate positive for any biomarker	874	71.4	92.3
	Prospective	Advanced adenoma	OC-Sensor (Eiken Chemical Company)	10 μg/g	1037	43.5	86.3	plasma	BCAT1/IKZF1	ctDNA methylation	1 replicate positive for any biomarker	1037	9.4	92.3
	Prospective	CRC	OC-Sensor (Eiken Chemical Company)	60 μg/g	933	63.6	94.3	plasma	BCAT1/IKZF1	ctDNA methylation	1 replicate positive for any biomarker	933	62.1	92.3
	Prospective	Stage I	OC-Sensor (Eiken Chemical Company)	60 μg/g	884	58.8	94.3	plasma	BCAT1/IKZF1	ctDNA methylation	1 replicate positive for any biomarker	884	41.2	92.3
	Prospective	Stage II	OC-Sensor (Eiken Chemical Company)	60 μg/g	892	68.0	94.3	plasma	BCAT1/IKZF1	ctDNA methylation	1 replicate positive for any biomarker	892	76.0	92.3
	Prospective	Stage III	OC-Sensor (Eiken Chemical Company)	60 μg/g	884	58.8	94.3	plasma	BCAT1/IKZF1	ctDNA methylation	1 replicate positive for any biomarker	884	58.8	92.3
	Prospective	Stage IV	OC-Sensor (Eiken Chemical Company)	60 μg/g	874	71.4	94.3	plasma	BCAT1/IKZF1	ctDNA methylation	1 replicate positive for any biomarker	874	71.4	92.3
	Prospective	Advanced adenoma	OC-Sensor (Eiken Chemical Company)	60 μg/g	1037	18.8	94.3	plasma	BCAT1/IKZF1	ctDNA methylation	1 replicate positive for any biomarker	1037	9.4	92.3
	Prospective	CRC	OC-Sensor (Eiken Chemical Company)	80 μg/g	933	59.1	95.6	plasma	BCAT1/IKZF1	ctDNA methylation	1 replicate positive for any biomarker	933	62.1	92.3
	Prospective	Stage I	OC-Sensor (Eiken Chemical Company)	80 μg/g	884	58.8	95.6	plasma	BCAT1/IKZF1	ctDNA methylation	1 replicate positive for any biomarker	884	41.2	92.3
	Prospective	Stage II	OC-Sensor (Eiken Chemical Company)	80 μg/g	892	60.0	95.6	plasma	BCAT1/IKZF1	ctDNA methylation	1 replicate positive for any biomarker	892	76.0	92.3
	Prospective	Stage III	OC-Sensor (Eiken Chemical Company)	80 μg/g	884	52.9	95.6	plasma	BCAT1/IKZF1	ctDNA methylation	1 replicate positive for any biomarker	884	58.8	92.3
	Prospective	Stage IV	OC-Sensor (Eiken Chemical Company)	80 μg/g	874	71.4	95.6	plasma	BCAT1/IKZF1	ctDNA methylation	1 replicate positive for any biomarker	874	71.4	92.3
	Prospective	Advanced adenoma	OC-Sensor (Eiken Chemical Company)	80 μg/g	1037	17.1	95.6	plasma	BCAT1/IKZF1	ctDNA methylation	1 replicate positive for any biomarker	1037	9.4	92.3

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Table 2 (*continued*)

			Faecal immunochemical test					Blood-based test						
Author	Study design	Disease state	Brand	Positivity threshold	N	Sn (%)	Sp (%)	Blood	Biomarker	Biomarker class	Positivity threshold	N	Sn (%)	Sp (%)
Tao [37]	Case-Control	CRC	RIDASCREEN Hb (R-Biopharm AG)	24 μg/g	284	65.7	97.7	serum	CRP	protein	2499.9 ng/mL	284	19.4	90.3
	Case-Control	Advanced adenoma	RIDASCREEN Hb (R-Biopharm AG)	24 μg/g	400	19.7	97.7	serum	CRP	protein	2499.9 ng/mL	400	4.9	90.3
	Case-Control	CRC	RIDASCREEN Hb (R-Biopharm AG)	24 μg/g	284	65.7	97.7	serum	sCD26	protein	133.5 ng/mL	284	0.0	97.2
	Case-Control	Advanced adenoma	RIDASCREEN Hb (R-Biopharm AG)	24 μg/g	400	19.7	97.7	serum	sCD26	protein	133.5 ng/mL	400	4.4	97.2
	Case-Control	CRC	RIDASCREEN Hb (R-Biopharm AG)	24 μg/g	284	65.7	97.7	serum	TIMP-1	protein	928.2 ng/mL	284	13.4	97.7
	Case-Control	Advanced adenoma	RIDASCREEN Hb (R-Biopharm AG)	24 μg/g	400	19.7	97.7	serum	TIMP-1	protein	928.2 ng/mL	400	7.1	97.7
Tsai [38]	Prospective case-control	CRC	ADVIA Centaur® Analyzer	100 ng/mL[a]	174	25.7	44.6	whole blood	EpCAMþ-CTC	CTC	23 EpCAMþ cells/ml	174	85.3	78.5
	Prospective case-control	CRC	ADVIA Centaur® Analyzer	100 ng/mL[a]	174	25.7	44.6	whole blood	PDPNþ-CTC	CTC	7 PDPNþ cells/ml	174	78.0	75.4
Werner [39]	prospective	CRC	RIDASCREEN Hb (R-Biopharm AG) (prior to 2009); OB Gold (Sentinel Diagnostics) (after 2009)	9.6 μg/g (RIDASCEEN); 42.5 μg/g (FOB Gold with frozen stool samples); 15.3 μg/g (FOB Gold with fresh stool samples)	922	78.0	96.0	serum	5 biomarker panel (CEA, ferritin, seprase, osteopontin, and anti-p53 antibody)	protein	pre-defined algorithm	922	39.0	96.0
	prospective	Advanced adenoma	RIDASCREEN Hb (R-Biopharm AG) (prior to 2009); OB Gold (Sentinel Diagnostics) (after 2009)	9.6 μg/g (RIDASCEEN); 42.5 μg/g (FOB Gold with frozen stool samples); 15.3 μg/g (FOB Gold with fresh stool samples)	1200	28.0	96.0	serum	5 biomarker panel (CEA, ferritin, seprase, osteopontin, and anti-p53 antibody)	protein	pre-defined algorithm	1200	8.0	96.0
Xie [40]	Case-Control	CRC	Colloidal gold immunochromatography with double antibody sandwich assay	200 ng/ml[a]	162	61.4	70.3	serum	CEA	protein	5.2 ng/mL	248	35.0	62.6
	Case-Control	CRC	Colloidal gold immunochromatography with double antibody sandwich assay	200 ng/ml[a]	162	61.4	70.3	serum	CA199	protein	39.0 U/mL	248	17.9	55.7
	Case-Control	CRC	Colloidal gold immunochromatography with double antibody sandwich assay	200 ng/ml[a]	162	61.4	70.3	serum	*SEPT9*	ctDNA methylation	Ct value < 35	248	61.8	89.6
	Case-Control	stage I	Colloidal gold immunochromatography with double antibody sandwich assay	200 ng/ml[a]	79	80.0	70.3	serum	*SEPT9*	ctDNA methylation	Ct value < 35	130	60.0	89.6
	Case-Control	stage II	Colloidal gold immunochromatography with double antibody sandwich assay	200 ng/ml[a]	97	56.5	70.3	serum	*SEPT9*	ctDNA methylation	Ct value < 35	161	52.8	89.6
	Case-Control	stage III	Colloidal gold immunochromatography with double antibody sandwich assay	200 ng/ml[a]	119	68.9	70.3	serum	*SEPT9*	ctDNA methylation	Ct value < 35	183	63.8	89.6
	Case-Control	stage IV	Colloidal gold immunochromatography with double antibody sandwich assay	200 ng/ml[a]	78	25.0	70.3	serum	*SEPT9*	ctDNA methylation	Ct value < 35	129	50.0	89.6

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Table 2 (*continued*)

Author	Study design	Disease state	Faecal immunochemical test					Blood-based test						
			Brand	Positivity threshold	N	Sn (%)	Sp (%)	Blood	Biomarker	Biomarker class	Positivity threshold	N	Sn (%)	Sp (%)
Ye [41]	Prospective	CRC	not reported	not reported	60	50.0	81.8	plasma	multi-panel	ctDNA methylation	PHF index ≥10	107	57.1	91.0
	Prospective	Advanced adenoma	not reported	not reported	60	38.5	81.8	plasma	multi-panel	ctDNA methylation	PHF index ≥10	107	13.3	91.0

CRC: colorectal cancer; Ct: cycle threshold; ctDNA: circulating tumour DNA; CTC: circulating tumour cells.
[a] Details not available to convert the positivity threshold to the recommended units of μg haemoglobin/g faeces.

2.2.4. Sensitivity and specificity of FIT and blood-based tests for CRC

The accuracy data is detailed in Table 2 and Supplementary Table 4. Sensitivity of FIT ranged from 17.1 % for advanced adenomas [36] up to 100 % for Stage IV CRC [29], with specificity as low as 42.3 % for all stages of CRC and as high as 98.3 % for advanced neoplasia [33]. For blood-based tests, the lowest sensitivity was 0 % for the sCD26 protein biomarker for all stages of CRC [37], while the highest sensitivity was 100 % for the protein biomarker TAP, and the highest specificity was observed for the multi-miRNA panel at 97.8 % [31]. The study with the highest FIT accuracy was Werner et al. (2015) [39], with a sensitivity of 78 % and specificity of 96 % for all stages of CRC (n = 992 participants). For the blood-based tests, measurement of TAP (protein) returned the highest accuracy of 96 % (sensitivity 100 %, specificity 86.7 %) amongst 99 participants. Given the large range of advancement of disease tested across the studies for both blood and FIT, herein we provide results from separate meta-analyses for detection of CRC at all stages, precancerous lesions, a paired-difference assessment of FIT versus blood-based tests for CRC, meta-analysis of early- and late-stage CRC, a sub-analysis of only those studies assessing the methylated *SEPT9* blood biomarker, and an assessment of the PPV value of the blood-based tests compared to FIT in studies that were of prospective design.

2.2.5. Meta-analysis of blood-based tests and FIT for the detection of CRC

There were a total of 37 unique blood tests across 15 studies that had any stage of CRC as the diagnostic outcome (Fig. 2a–b). Blood biomarkers included CTCs (n = 1), DNA methylation (n = 9), miRNA (n = 1), mRNA (n = 2) and proteins (n = 5). For FIT (Fig. 2c–d), there were 15 studies that assessed any stage of CRC as the outcome, with one study [36] assessing three f-Hb positivity thresholds (10, 60 and 80 μg/g Hb). The pooled sensitivity and specificity for CRC of blood-based tests were 57 % (95 % CI 42–71 %) and 89 % (95 % CI 85–93 %), respectively. The pooled sensitivity and specificity for FIT were 65 % (95 % CI 56–74 %) and 82 % (95 % CI 72–91 %), respectively.

2.2.6. Meta-analysis of blood-based tests and FIT for the detection of advanced precursor lesions, early-stage and late-stage CRC

There were 29 unique blood-based tests tested across 10 studies for advanced precancerous lesions, including advanced adenomas and/or sessile serrated lesions as the outcome (Fig. 3a). Blood biomarker types included DNA methylation (n = 5), miRNA (n = 1), and proteins (n = 4). Fifteen assessments for FIT accuracy across 11 studies were assessed for detection of advanced precancerous lesions (Fig. 3b), with two studies included more than once due to the assessment of multiple FIT thresholds [26,36], as well as analysing advanced adenomas and sessile serrated lesions separately [26]. The pooled sensitivity for precancerous lesions for blood-based tests versus FIT were 22 % (95 % CI 8–36 %) versus 29 % (95 % CI 24–34 %), respectively.

For early- and late-stage CRC there were 6 unique blood tests from 6 studies (Fig. 3c and e) and 8 unique FIT measurements from 6 studies (Fig. 3d and f). The pooled sensitivity of blood-based tests for detecting early- and late-stage CRC were 54 % (95 % CI 32–76 %) and 60 % (95 % CI 35–85 %), respectively. For FIT, the pooled sensitivity for detecting early- and late-stage CRC were 64 % (95 % CI 50–78 %) and 79 % (95 % CI 66–92 %), respectively.

2.2.7. Paired-difference meta-analysis for FIT versus blood-based tests

For a direct head-to-head comparison of the blood tests versus FIT, a paired-difference meta-analysis was conducted on the sensitivity and specificity for CRC and advanced precancerous lesions, presented as SROC curves (Fig. 4). FIT had a significantly higher sensitivity for CRC (Fig. 4a; $p < 0.01$) and advanced precancerous lesions (Fig. 4b; $p < 0.01$) when compared to blood tests. There were no significant differences in specificity between FIT and blood-based tests for either CRC ($p = 0.085$; Fig. 4a) or precancerous lesions ($p = 0.979$; Fig. 4b).

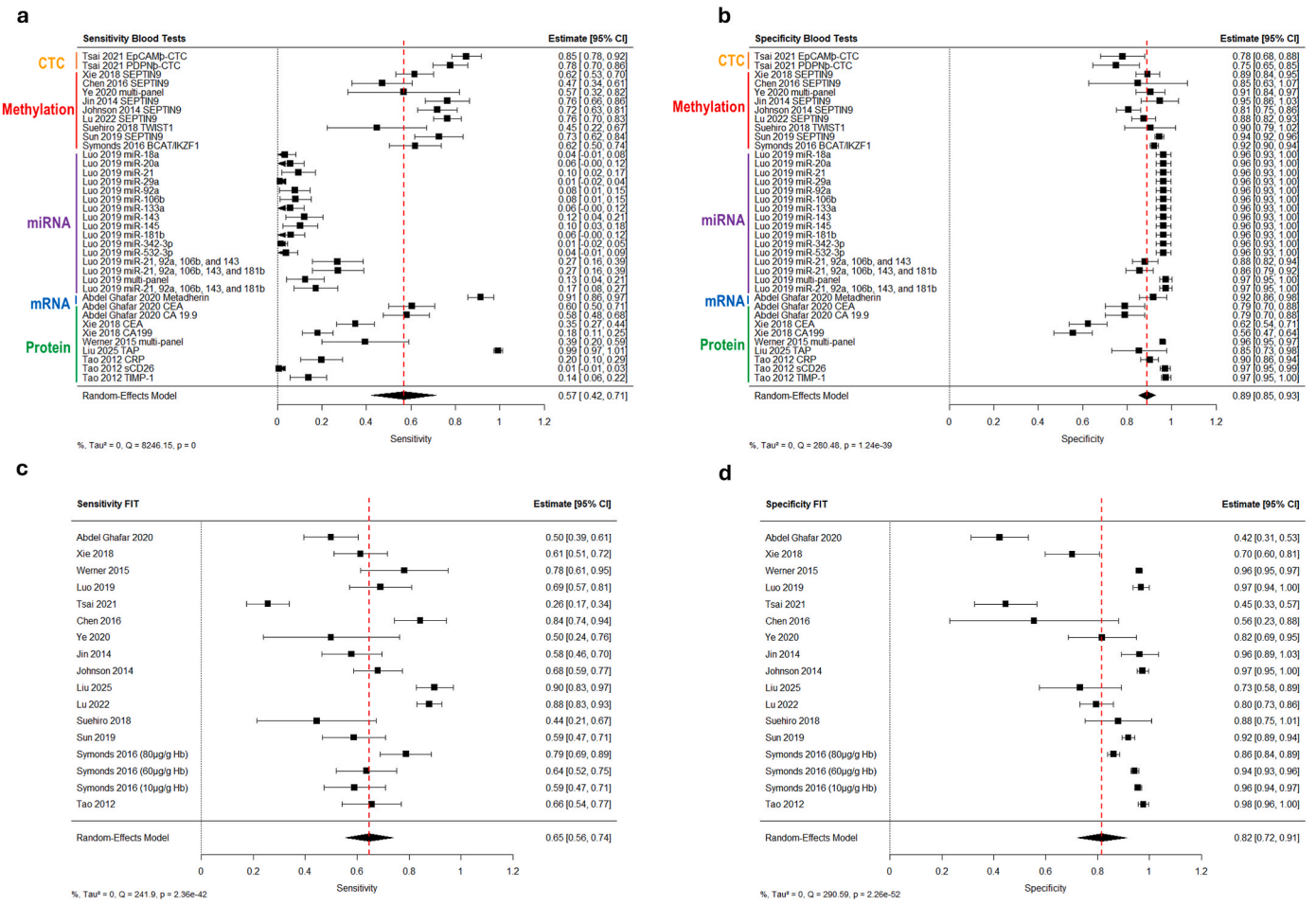


Fig. 2. Multivariable REML meta-analyses of blood test sensitivity (a) and specificity (b) and FIT sensitivity (c) and specificity (d) for detection of any stage of CRC. CI; confidence interval, CTC; circulating tumour cell.

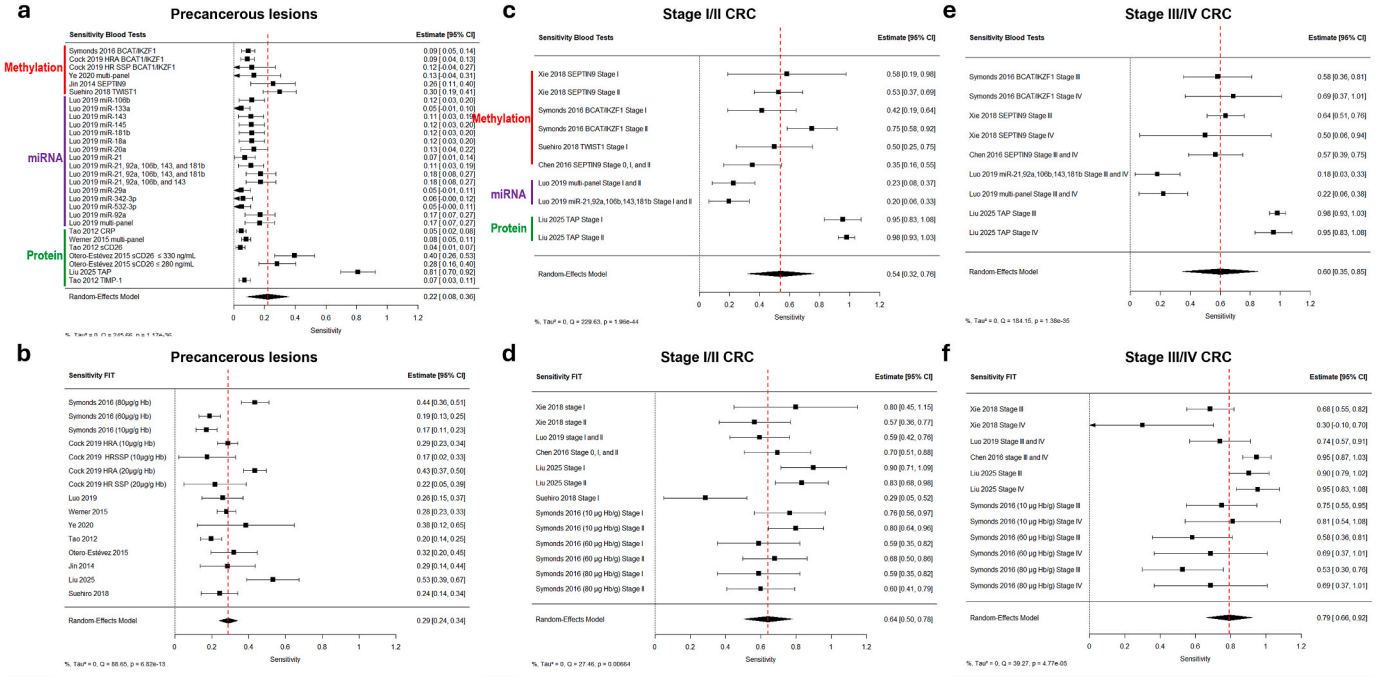


Fig. 3. Multivariable REML meta-analyses of blood test sensitivity and specificity for advanced precancerous lesions (a–b), early-stage CRC (c–d) and late-stage CRC (e–f).

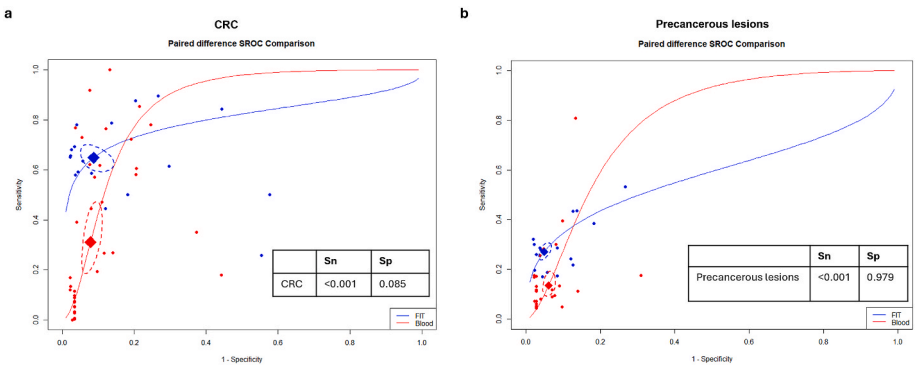


Fig. 4. Pairwise meta-analysis for FIT versus blood for the detection of CRC (a) and advanced precancerous lesions (b). Wilcoxon signed rank test with continuity correction. CI; confidence interval, Sn; sensitivity, Sp; specificity, SROC; summary receiver operating characteristic.

2.2.8. *Meta-analysis of methylated SEPTIN9 for the detection of CRC*

The *SEPT9* methylation assay had the most included published studies in this review and is the only blood test reviewed here that has FDA approval (Epi proColon). The pooled sensitivity and specificity for methylated *SEPT9* for CRC was 69 % (95 % CI 60–77 %) and 90 % (95 % CI 85–95 %), respectively (Fig. 5a–b). For FIT, the pooled sensitivity and specificity was 70 % (95 %CI 59–81 %) and 85 % (95 % CI 75–96 %), respectively (Fig. 5c–d). When comparing *SEPT9* methylation and FIT in a paired-difference meta-analysis, there were no significant differences in sensitivity or specificity between the two tests (Fig. 5e, $p = 0.844$, $p = 0.281$).

2.2.9. *Meta-analysis of the PPV for blood-based tests and FIT to detect CRC in prospective studies*

When limiting to the five studies with a prospective design to calculate PPV, the pooled PPV for FIT (Fig. 6a) versus blood (Fig. 6b) were not different, at 66 % (95 % CI 60–72 %) versus 60 % (95 % CI 32–88 %), respectively.

2.2.10. *Discussion*

This is the first systematic review which assesses studies that directly compare blood-based tests to FIT. We found that very few studies had performed this comparison, and not all used a quantitative FIT. As discussed earlier, it is recommended to compare new blood-based tests to a brand of FIT that has undergone robust validation [18]. The data have shown that blood-based tests have a lower pooled sensitivity compared to FIT for both CRC and precancerous lesions. While we have also assessed PPV, limiting it to prospective studies, the studies are not restricted to a true screening population, with many studies enriching for CRC cases. A further limitation is that there were several different FIT brands used, and multiple thresholds applied from 3.5 μg/g to 80 μg/g. While lower thresholds have higher sensitivity for neoplasia detection, specificity is impacted with more false positives. Our SROC analysis showed that while overall specificity was not different between FIT and blood tests, sensitivity was significantly higher for both CRC and advanced precancerous lesions with FIT. More work is needed in this field, with benefits to be gained from further discovery of blood tests with high sensitivity for precancerous lesions.

3. Logistics of CRC screening program implementation

When implementing a new CRC screening test into routine practice, there are several logistical considerations across the pre-analytical, analytical (i.e. laboratory), and post-analytical stages. These not only influence the accuracy of the test but also the program effectiveness.

3.1. *Pre-analytical considerations*

We have previously described how the accuracy of FIT can be impacted by collection method, stool consistency, blood type, and transport conditions (temperature and duration to the testing laboratory) [42]. The same evaluation will be necessary for blood-based tests

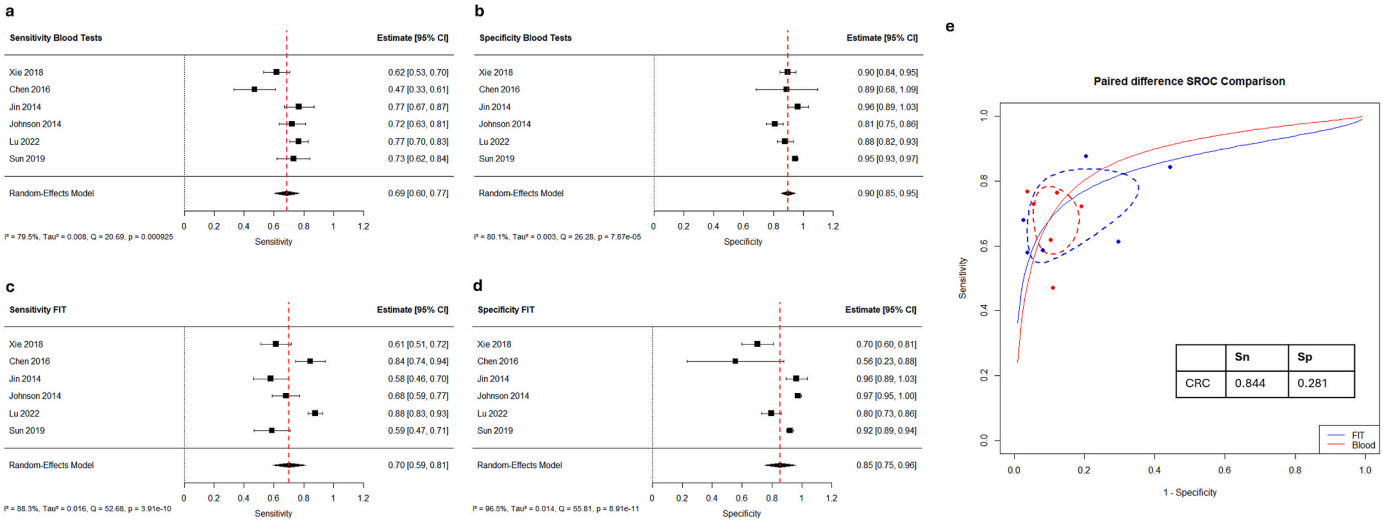


Fig. 5. Bivariate REML meta-analyses of methylated *SEPTIN 9* sensitivity (a) and specificity (b), and FIT sensitivity (a) and specificity (b) for CRC from studies assessing *SEPTIN 9* accuracy. CI; confidence interval.

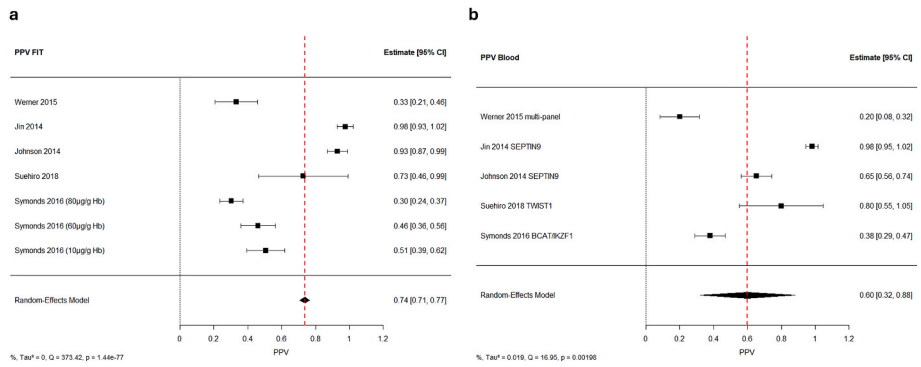


Fig. 6. Bivariate REML meta-analyses for the positive predictive (PPV) of FIT (a) and multivariable REML metanalysis for PPV of blood tests (b) for CRC in studies with a prospective design. CI; confidence interval.

to optimise and standardise collection and processing techniques. For example, does blood sampling need to be at a certain time of day or after fasting; and how should samples be transported to the laboratory and processed? Delayed processing can cause cellular degradation of the blood sample, releasing genomic DNA and diluting tumour-derived DNA [43]. cfDNA will also be degraded by nucleases, and therefore samples should be frozen or extracted soon after plasma separation. Large variability across laboratories for cfDNA pre-analytical processes has been reported [43].

3.2. Analytical considerations

A critical factor in program implementation of any diagnostic test is the analytical robustness and reproducibility of the test. For blood-based tests there have been proposals for materials that could be used for quality control measures [44]. Even with FIT there is limited standardisation and harmonisation with external quality assessment schemes, but recommendations exist [45]. One such approach for standardisation for FIT is that all positivity thresholds be expressed as µg Hb per gram of stool [46], as this makes results comparable between different brands of tests. Such consistency may be harder to achieve with blood-based tests that measure various molecular biomarkers (e.g. protein, ctDNA) that have different outputs, and those that rely on scores from algorithms based on biomarker detection parameters rather than concentrations of individual biomarkers.

If a blood-based test has an adjustable positivity threshold (ideal, but maybe not possible), each country would need to determine an appropriate cut-off to meet the goals of their screening. An adjustable threshold is one advantage of FIT, allowing colonoscopy workloads to be managed, particularly during times of constraint (such as during the COVID pandemic [47]), and in facilitating personalisation of screening intervals [48] and risk stratification for sex [49]. A recent international survey showed that for FIT, countries that selected a higher positivity threshold are guided by colonoscopy demand, while lower thresholds focus on detection of precancerous lesions to reduce CRC incidence [16]. Other considerations are the number of test replicates. FIT has a higher sensitivity with a two-sample collection process [50], while the blood-based methylated *SEPT9* test has a higher detection rate with 1/3 replicates deemed positive compared with 2/3 replicates [51]. While test adaptability allows countries to tailor their screening strategies, assay standardisation remains crucial for comparability and quality assurance.

3.3. Post-analytical considerations

FIT has known sources of false positives and negatives including sex, age, and some medications [52], while blood-based cfDNA levels can be influenced by physical activity, fasting status, circadian rhythm, age, and chronic inflammatory conditions [53]. For the methylated *SEPT9* test, arthritis was associated with an increased false negative rate, while arteriosclerosis was associated with an increased false positive rate [54]. Furthermore, methylated *SEPT9* can also detect cancers other than CRC [40,55–57]. This raises the question of the clinical pathway after a positive screening test but negative colonoscopy. After a positive FIT/-negative colonoscopy, upper gastroscopy is not recommended [58], but it is not clear whether this will also be the case after a positive blood-based test.

Finally, longitudinal screening (i.e. repeated testing) is essential to ensure program effectiveness through detecting any missed or growing lesions. FIT is provided every 1–2 years after a negative result [5], while the CMS will cover a blood-based CRC screening test once every 3 years. FIT provides quantitative results, allowing for risk stratification over time, with rising faecal Hb concentrations associated with future neoplasia risk [59]. As the FDA approved blood-based tests provide qualitative results (normal/abnormal), such longitudinal monitoring will not be possible.

4. Consideration of preferences for CRC screening modality

Test preferences play a significant role in CRC screening uptake. Many individuals, particularly those without faecal-based screening experience, prefer the idea of blood-based tests due to reported faecal aversion [13] and perceptions of greater simplicity, convenience, and familiarity [13,60–62]. Offering blood-based CRC screening may increase engagement, particularly among those hesitant to participate [61]. Other individuals have reported greater accountability with attending a scheduled blood draw than completing FIT at home [60,62], however, the need to travel for in-person blood collection can counteract these benefits [62,63].

In contrast, individuals with prior FIT experience are more likely to prefer stool-based tests [64], considering them more accurate [60] and blood collection less convenient [65]. Across studies, test accuracy is the primary driver of choice [13,61], influenced by sociodemographic factors. Older adults, White individuals, and those with higher education or socioeconomic status generally prefer tests with higher accuracy [66, 67]. However, test preferences are not always aligned with previous behaviours and many still prefer a blood test when given a choice [65].

5. Participation with screening tests

FIT participation varies between program type (organised versus opportunistic) and setting (general population versus clinical settings). For an effective program, European guidelines recommend an uptake of at least 65 % [68], while the US guidelines recommend 80 % [11]. FIT uptake is below these targets in many countries [69], with a review demonstrating that FIT uptake ranged from 7 % to 96 %, with uptake of 55 % in Asia Pacific, 53 % in Europe, 46 % in North America, and 90 % in South America [69]. The key determinants of a higher FIT uptake

include a lower number of FIT samples needed (one versus two), subsequent rounds of participation (subsequent versus first), and older age [69,70]. Key barriers include low awareness, fear of cancer, test disgust, limited healthcare access, and sociodemographic disparities [71–73]. Strategies such as reminders, culturally tailored programs, and simplified test delivery have shown some success at improving uptake [72,74].

Blood-based tests may offer an alternative pathway to improve uptake [74], allowing individuals to select the test that suits them [75], with clinical settings including physician endorsement [76] and convenience [77] supporting higher participation. Studies in Germany and the US report blood test uptake of 83 % [87] and 93.5 % [84] when unscreened individuals within a healthcare setting get a choice between blood and FIT. However, blood test uptake is lower when invitations are mailed or phone-based. One such study found that a blood test offer only increased overall screening participation by 20.4 % (to 30.5 %) [78], with other studies showing lower increases such as blood test uptake of 5 % (overall participation of 17.1 %) [79] and 3.8 % (overall participation of 36 %) [80]. In addition, intention cannot guarantee action, with a study showing that uptake is only 57.5 % in individuals requesting information about the blood test [80]. Follow-up adherence after a positive blood test remains a concern, with colonoscopy rates lower than those following a positive FIT. European guidelines recommend >90 % colonoscopy follow-up [81], yet real-world adherence is less (76 % after FIT [82]; 25–50 % after blood-based tests [78,83]).

Our discussion of participation has focussed on single rounds of screening, but for program effectiveness, screening should be completed at regular intervals There has been limited research into long-term adherence with blood-based tests in comparison to FIT, but our own study demonstrated no difference to participation in the second round of screening for FIT (66 %) and blood tests (61 %) [84]. Again, blood test completion would likely be more successful if it was administered by the primary care physician.

6. Cost effectiveness of screening

With sensitivity and accuracy ascertained, efficient allocation of health care resources requires focus on the cost and effectiveness amidst all other competing testing modalities. If blood-based tests are superior in accuracy, are they also superior in cost for every CRC avoided? Who pays for this test and how much are they willing to pay?

Table 3 summarises some of the costs that have been applied in analysis comparing blood tests to faecal-based tests and colonoscopy in the last five years.

Table 3
Costs of faecal immunochemical tests, blood-based tests and colonoscopy applied in cost effectiveness modelling studies.

Study	Cost		
	Faecal immunochemical test	Blood-based test	Colonoscopy
Barre et al. (2020) and Benamouzig et al. (2021) [85,86]	Euro 14.34	Euro 125.13	Without adenoma removal Euro 806.44 With adenoma removal Euro 1191.6
Ladabaum et al. (2024) [87]	USD 18.05	USD 1495	
Aziz et al. (2023) [88]	USD 52	USD 949	Colonoscopy without polypectomy US 1119 Colonoscopy with polypectomy US 1366
Nascimento de lima et al. (2025) and van den Puttelaar et al. (2024) [89,90]	USD 25.53	USD 895	Without lesion removal USD 1427 With lesion removal USD 1889.78

Although a blood test is cost effective compared to no screening for CRC screening, it is not cost effective compared to FIT [85,86,91,92]. Recent analysis comparing faecal tests to blood tests (following current CMS coverage decisions), for CRC screening in the US setting [89,90] it has been demonstrated that even with 100 % adherence to the triennial blood test, annual FIT is more effective at reducing CRC incidence and mortality and more cost effective. Ladabaum et al. (2024) argued that to achieve reduction in CRC incidence and mortality that is comparable to annual FIT (40 % adherence/participation), adherence with the blood test would have to be 1.8 and 1.5-fold higher respectively [93].

When assessing cost effectiveness of the Australian National Bowel Cancer screening program, it was found that compared to ctDNA blood tests, FIT screening every 2 years was cost-effective in reducing CRC incidence and mortality at participation adherence rates ranging from 29 % to 100 % [94]. At low participation, 2 yearly FIT reduced CRC incidence and mortality by 23 % and 36 %, while at 100 % participation and adherence these outcomes reduced by 51 % and 74 % respectively [94]. Conversely, with a 2-yearly blood-based test, the reduction would only be 13 % and 28 % at 30 % participation compared to 37 % and 66 % at 100 % participation. It has been suggested that the ability of the blood test to detect advanced precancerous lesions would have greater impact on cost effectiveness compared to sensitivity for CRC [93,94]. The blood test with a sensitivity of 74–95 % for CRC and 70–80 % for precancerous lesions would be cost-effective compared to annual FIT in the US (with a willingness to pay threshold of $100,000).

Nascimento de Lima et al. (2024) explored cost effectiveness if the blood test was non-inferior to FIT in cost, sensitivity for CRC and advanced adenomas and offered at similar intervals [95]. Compared to annual FIT, an annual blood test with 92 % and 50 % sensitivity for CRC and advanced adenoma (the best accuracy assumptions) respectively had similar effectiveness but was not cost-effective owing to the higher cost of the blood test (US$500). This analysis established a cost threshold of $125 for an annual blood test if the sensitivity for advanced adenomas is higher than 40 % and above 83 % for CRC [95]. Another study showed that a blood-based test with superior sensitivity and specificity (100 %) to FIT (53 % and 100 % sensitivity for large adenomas and CRC, respectively, and 100 % specificity) would be cost effective if its cost did not exceed 7-fold the cost of FIT [96]. Puttelaar et al. (2024) showed that even at a cost of zero for the blood test, annual FIT screening was less costly in the US [90].

Collectively, these studies suggest that while blood-based tests could provide an alternative for those unwilling to complete FIT, their widespread implementation in screening programs is not currently justified based on cost-effectiveness. Therefore efforts should be focused on improving FIT participation to achieve even greater effectiveness and cost effectiveness of CRC screening compared to improving participation in the blood test to 100 %.

7. Future directions

While interest in blood-based screening tests is increasing, particularly due to their potential to increase participation, several limitations remain. Few blood biomarkers have been directly compared to quantitative FIT, and there have been multiple FIT brands and positivity thresholds applied across studies, and so caution should be made with interpretation of comparisons. It is also clear that blood-based tests are not yet cost effective and have limited sensitivity for advanced precancerous lesions and early-stage CRC, which are essential features for effective screening and cancer prevention. Evidence that a test can detect CRC does not ensure that it can reduce mortality rates [97], especially as blood tests are more sensitive for late stage disease.

Detection and removal of precancerous lesions is needed to reduce

CRC incidence, yet most blood tests show poor sensitivity, even lower than that of FIT. The CMS covers blood tests with sensitivity of at least 74 % for detection of CRC and specificity of at least 90 %, but does not specify the sensitivity requirement for precancerous lesion detection [19]. The two blood tests that are approved by the FDA (Shield and Epi proColon; the former has not been included in the systematic review as there are no published studies including comparison to FIT), are reported to have a sensitivity for CRC of 83 % and 71 %, respectively, and specificity of 90 % and 92 % [14,26]. Precancerous lesion sensitivity is 13 % for Shield and 23 % for methylated *SEPT9* [14,15]. The low precancerous lesion sensitivity highlight that blood-based tests are not yet a substitute for FIT for primary screening.

Blood tests still have a role in screening. They could be utilised in younger adults (45-50y) where screening has now been recommended in several countries [98,99]. FIT participation is generally lowest in the youngest age groups [100], with these individuals more likely to comply with a technology that they are more familiar with, such as blood testing [101]. Similarly, in older populations who are still eligible for screening (75-80y), stool tests may be impractical due to benign colorectal bleeding conditions or physical limitations, making blood-based screening a more feasible option. Blood tests also have a proven role in providing a screening technology for individuals who are unwilling or unable to complete FIT [75]. Medical conditions include cognitive decline, vision or mobility issues may cause barriers to FIT completion, and therefore offering a blood-based option allows these individuals to engage in preventive care [75].

Our goal should be to prevent cancers, and not just detect them. Therefore, efforts should focus on increasing engagement in CRC screening with FIT. Ongoing evaluation is needed as new blood-based technologies are adopted. Ultimately, effective screening must aim not just to detect cancers, but to prevent them. Blood is not yet ready to be the new stool test.

Practice points

- Relatively few studies have made a direct comparison of diagnostic accuracy of blood-based tests to FIT. Methylated *SEPT9* has had the most evaluation.
- The blood-based tests had a pooled sensitivity for CRC of 36 % (95 % CI 26–46 %) which was lower than FIT which had a sensitivity of 65 % (95 % CI 57–73 %). The pooled specificity was not significantly different between the tests (blood-based tests: 91 % (95 % CI 88–93 %); FIT: 83 % (95 % CI 75–91 %)).
- Blood-based tests have a lower pooled sensitivity for early-stage cancer (55 % (95 % CI 37–74 %)) and for precancerous lesions (15 % (95 % CI 10–20 %)). In comparison, FIT sensitivity is 68 % (95 % CI 59–76 %) and 29 % (95 % CI 23–34 %) for early-stage cancer and for precancerous lesions respectively.
- Blood-based CRC screening is currently not cost effective.
- CRC screening with blood-based tests may improve participation in individuals that are unwilling or unable to complete other screening tests, but participation results vary between studies, with greater participation in studies conducted within healthcare settings.

Research agenda

- Behavioural studies are essential as further evidence is needed to confirm the uptake, adherence with diagnostic colonoscopy, and long-term repeat participation in blood-based CRC screening.
- Further research is needed to identify blood-based markers with high sensitivity for advanced precancerous lesions, including advanced adenomas and sessile serrated lesions.
- Additional data are needed on diagnostic accuracy of blood-based biomarkers across different CRC stages, particularly for early-stage disease.
- Evaluation is required to determine whether current blood biomarkers used for CRC screening also detect extracolonic cancers, and if so, to guide appropriate follow-up diagnostics.

CRediT authorship contribution statement

Erin L. Symonds: Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Maddison Dix:** Writing – review & editing, Writing – original draft, Conceptualization. **Molla M. Wassie:** Writing – review & editing, Writing – original draft, Conceptualization. **Norma B. Bulamu:** Writing – review & editing, Writing – original draft, Conceptualization. **Graeme P. Young:** Writing – review & editing, Conceptualization. **Jean M. Winter:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization.

Role of the funding source

The funders had no role in the collection, analysis or interpretation of the data presented.

Declaration of competing interest

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Appendix A. Supplementary data

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References

[1] Bray F, Laversanne M, Sung H, Ferlay J, Siegel RL, Soerjomataram I, et al. Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA Cancer J Clin 2024;74(3):229–63.
[2] Nguyen LH, Goel A, Chung DC. Pathways of colorectal carcinogenesis. Gastroenterology 2020;158(2):291–302.
[3] Mandel JS, Church TR, Bond JH, Ederer F, Geisser MS, Mongin SJ, et al. The effect of fecal occult-blood screening on the incidence of colorectal cancer. N Engl J Med 2000;343(22):1603–7.
[4] Young GP, Symonds EL, Allison JE, Cole SR, Fraser CG, Halloran SP, et al. Advances in fecal occult blood tests: the FIT revolution. Dig Dis Sci 2015;60(3):609–22.
[5] Schreuders EH, Ruco A, Rabeneck L, Schoen RE, Sung JJ, Young GP, et al. Colorectal cancer screening: a global overview of existing programmes. Gut 2015;64(10):1637–49.
[6] Grobbee EJ, Wisse PHA, Schreuders EH, van Roon A, van Dam L, Zauber AG, et al. Guaiac-based faecal occult blood tests versus faecal immunochemical tests for colorectal cancer screening in average-risk individuals. Cochrane Database Syst Rev 2022;6(6):CD009276.
[7] Chiu HM, Jen GH, Wang YW, Fann JC, Hsu CY, Jeng YC, et al. Long-term effectiveness of faecal immunochemical test screening for proximal and distal colorectal cancers. Gut 2021.
[8] Zorzi M, Urso EDL. Impact of colorectal cancer screening on incidence, mortality and surgery rates: evidences from programs based on the fecal immunochemical test in Italy. Dig Liver Dis : Offic J Italian Soc Gastroenterol Italian Assoc Study Liver 2023;55(3):336–41.

[9] Chiu HM, Chen SL, Yen AM, Chiu SY, Fann JC, Lee YC, et al. Effectiveness of fecal immunochemical testing in reducing colorectal cancer mortality from the one million Taiwanese screening program. Cancer 2015;121(18):3221–9.
[10] Vart G, Banzi R, Minozzi S. Comparing participation rates between immunochemical and guaiac faecal occult blood tests: a systematic review and meta-analysis. Prev Med 2012;55(2):87–92.
[11] Wender RC, Doroshenk M, Brooks D, Hotz J, Smith RA. Creating and implementing a national public health campaign: the American cancer society's and national colorectal cancer roundtable's 80% by 2018 initiative. Am J Gastroenterol 2018;113(12):1739–41.
[12] Worthley DL, Cole SR, Esterman A, Mehaffey S, Roosa NM, Smith A, et al. Screening for colorectal cancer by faecal occult blood test: why people choose to refuse. Intern Med J 2006;36(9):607–10.
[13] Osborne JM, Flight I, Wilson CJ, Chen G, Ratcliffe J, Young GP. The impact of sample type and procedural attributes on relative acceptability of different colorectal cancer screening regimens. Patient Prefer Adherence 2018;12:1825–36.
[14] Nian J, Sun X, Ming S, Yan C, Ma Y, Feng Y, et al. Diagnostic accuracy of methylated SEPT9 for blood-based colorectal cancer detection: a systematic review and meta-analysis. Clin Transl Gastroenterol 2017;8(1):e216.
[15] Chung DC, Gray DM, Singh H, Issaka RB, Raymond VM, Eagle C, et al. A cell-free DNA blood-based test for colorectal cancer screening. N Engl J Med 2024;390(11):973–83.
[16] Young GP, Benton SC, Bresalier RS, Chiu HM, Dekker E, Fraser CG, et al. Fecal immunochemical test positivity thresholds: an international survey of population-based screening programs. Dig Dis Sci 2024.
[17] Young GP, Senore C, Mandel JS, Allison JE, Atkin WS, Benamouzig R, et al. Recommendations for a step-wise comparative approach to the evaluation of new screening tests for colorectal cancer. Cancer 2016;122(6):826–39.
[18] Bresalier RS, Senore C, Young GP, Allison J, Benamouzig R, Benton S, et al. An efficient strategy for evaluating new non-invasive screening tests for colorectal cancer: the guiding principles. Gut 2023;72(10):1904.
[19] Centers for Medicare and Medicaid Services. Screening for colorectal cancer - blood-based biomarker tests - decision memo 2021 [Available from: https://www.cms.gov/medicare-coverage-database/view/ncacal-decision-memo.aspx?proposed=N&NCAId=299; 2021.
[20] Meester RG, Doubeni CA, Zauber AG, Goede SL, Levin TR, Corley DA, et al. Public health impact of achieving 80% colorectal cancer screening rates in the United States by 2018. Cancer 2015.
[21] Baldacchini F, Bucchi L, Giuliani O, Mancini S, Ravaioli A, Vattiato R, et al. Results of compliant participation in five rounds of fecal immunochemical test screening for colorectal cancer. Clin Gastroenterol Hepatol : Offi Clin Practice J Am Gastroenterol Assoc 2021;19(11):2361–9.
[22] Pepe MS, Etzioni R, Feng Z, Potter JD, Thompson ML, Thornquist M, et al. Phases of biomarker development for early detection of cancer. J Natl Cancer Inst 2001;93(14):1054–61.
[23] Whiting PF, Rutjes AW, Westwood ME, Mallett S, Deeks JJ, Reitsma JB, et al. QUADAS-2: a revised tool for the quality assessment of diagnostic accuracy studies. Ann Intern Med 2011;155(8):529–36.
[24] Abdel Ghafar MT, Gharib F, Abdel-Salam S, Elkhouly RA, Elshora A, Shalaby KH, et al. Role of serum metadherin mRNA expression in the diagnosis and prediction of survival in patients with colorectal cancer. Mol Biol Rep 2020;47(4):2509–19.
[25] Chen CH, Yan SL, Yang TH, Chen SF, Yeh YH, Ou JJ, et al. The relationship between the methylated Septin-9 DNA blood test and stool occult blood test for diagnosing colorectal cancer in Taiwanese people. J Clin Lab Anal 2017;31(1).
[26] Cock C, Anwar S, Byrne SE, Meng R, Pedersen S, Fraser RJL, et al. Low sensitivity of fecal immunochemical tests and blood-based markers of DNA hypermethylation for detection of sessile serrated adenomas/polyps. Dig Dis Sci 2019;64(9):2555–62.
[27] Jin P, Kang Q, Wang X, Yang L, Yu Y, Li N, et al. Performance of a second-generation methylated SEPT9 test in detecting colorectal neoplasm. J Gastroenterol Hepatol 2015;30(5):830–3.
[28] Johnson DA, Barclay RL, Mergener K, Weiss G, König T, Beck J, et al. Plasma Septin9 versus fecal immunochemical testing for colorectal cancer screening: a prospective multicenter study. PLoS One 2014;9(6).
[29] Liu X, Zhao G, Mao W, Li Q, Liao J, He G. Clinical significance of fecal Syndecan-2 gene methylation combined with blood tumor abnormal protein detection in the diagnosis of colorectal cancer and precancerous lesions. Clin Biochem 2025;136.
[30] Lu DC, Zhang QF, Li L, Luo XK, Liang B, Lu YH, et al. Methylated Septin9 has moderate diagnostic value in colorectal cancer detection in Chinese population: a multicenter study. BMC Gastroenterol 2022;22(1).
[31] Luo X, Wu Y, Ji M, Zhang S. Combined plasma microRNA and fecal occult blood tests in early detection of colorectal cancer. Clin Lab 2019;65(5):733–41.
[32] Otero-Estévez O, De Chiara L, Barcia-Castro L, de la Cadena MP, Rodríguez-Berrocal FJ, Cubiella J, et al. Evaluation of serum nucleoside diphosphate kinase A for the detection of colorectal cancer. Sci Rep 2016;6.
[33] Otero-Estévez O, De Chiara L, Rodríguez-Berrocal FJ, de la Cadena MP, Cubiella J, Castro I, et al. Serum sCD26 for colorectal cancer screening in family-risk individuals: comparison with faecal immunochemical test. Br J Cancer 2015;112(2):375–81.
[34] Suehiro Y, Hashimoto S, Higaki S, Fujii I, Suzuki C, Hoshida T, et al. Blood free-circulating DNA testing by highly sensitive methylation assay to diagnose colorectal neoplasias. Oncotarget 2018;9(24):16974–87.
[35] Sun J, Fei F, Zhang M, Li Y, Zhang X, Zhu S, et al. The role of mSEPT9 in screening, diagnosis, and recurrence monitoring of colorectal cancer. BMC Cancer 2019;19(1).
[36] Symonds EL, Pedersen SK, Baker RT, Murray DH, Gaur S, Cole SR, et al. A blood test for methylated BCAT1 and IKZF1 vs. a fecal immunochemical test for detection of colorectal neoplasia. Clin Transl Gastroenterol 2016;7(1):E137.
[37] Tao S, Haug U, Kuhn K, Brenner H. Comparison and combination of blood-based inflammatory markers with faecal occult blood tests for non-invasive colorectal cancer screening. Br J Cancer 2012;106(8):1424–30.
[38] Tsai WS, Hung WS, Wang TM, Liu H, Yang CY, Wu SM, et al. Circulating tumor cell enumeration for improved screening and disease detection of patients with colorectal cancer. Biomed J 2021;44(6):S190–200.
[39] Werner S, Krause F, Rolny V, Strobl M, Morgenstern D, Datz C, et al. Evaluation of a 5-Marker blood test for colorectal cancer early detection in a colorectal cancer screening setting. Clin Cancer Res 2016;22(7):1725–33.
[40] Xie L, Jiang X, Li Q, Sun Z, Quan W, Duan Y, et al. Diagnostic value of methylated Septin9 for colorectal cancer detection. Front Oncol 2018;8(JUL).
[41] Ye X, Chai H, Huang C, Liu M, Deng T. Can next-generation sequencing replace fecal immunochemical tests or ct in the screening of colorectal cancer and advanced adenoma? J Coll Physician Surg Pak 2020;30(9):940–5.
[42] Benton SC, Symonds E, Djedovic N, Jones S, Deprez L, Kocna P, et al. Faecal immunochemical tests for haemoglobin: analytical challenges and potential solutions. Clin Chim Acta 2021;517:60–5.
[43] Shin S, Woo HI, Kim JW, M DY, Lee KA. Clinical practice guidelines for pre-analytical procedures of plasma epidermal growth factor receptor variant testing. Ann Lab Med 2022;42(2):141–9.
[44] Devonshire A, Jones G, Gonzalez AF, Kofanova O, Trouet J, Pinzani P, et al. Interlaboratory evaluation of quality control methods for circulating cell-free DNA extraction. Nat Biotechnol 2023;78:13–21.
[45] Rubeca T, Rapi S, Deandrea S, Malaspina M, Passamonti BU, Grassi E, et al. Guidance for faecal occult blood testing: quantitative immunochemical method (FIT-HB) in colorectal cancer screening programmes. Epidemiol Prev 2017;41(1):1–31. 5-6.
[46] Fraser CG, Allison JE, Halloran SP, Young GP. Expert working group on fecal immunochemical tests for hemoglobin CCSCWEO. A proposal to standardize reporting units for fecal immunochemical tests for hemoglobin. J Natl Cancer Inst 2012;104(11):810–4.
[47] Loveday C, Sud A, Jones ME, Broggio J, Scott S, Gronthound F, et al. Prioritisation by FIT to mitigate the impact of delays in the 2-week wait colorectal cancer referral pathway during the COVID-19 pandemic: a UK modelling study. Gut 2020.
[48] Breekveldt ECH, Toes-Zoutendijk E, de Jonge L, Spaander MCW, Dekker E, van Kemenade FJ, et al. Personalized colorectal cancer screening: study protocol of a mixed-methods study on the effectiveness of tailored intervals based on prior f-Hb concentration in a fit-based colorectal cancer screening program (PERFECT-FIT). BMC Gastroenterol 2023;23(1):45.
[49] Blom J, Löwbeer C, Elfström KM, Sventelius M, Öhman D, Saraste D, et al. Gender-specific cut-offs in colorectal cancer screening with FIT: increased compliance and equal positivity rate. J Med Screen 2019;26(2):92–7.
[50] Young GP, Woodman RJ, Ang FLI, Symonds EL. Both sample number and test positivity threshold determine colonoscopy efficiency in detection of colorectal cancer with quantitative fecal immunochemical tests. Gastroenterology 2020;159(4):1561–3 e3.
[51] Song L, Jia J, Yu H, Peng X, Xiao W, Gong Y, et al. The performance of the mSEPT9 assay is influenced by algorithm, cancer stage and age, but not sex and cancer location. J Cancer Res Clin Oncol 2017;143(6):1093–101.
[52] de Klerk CM, Vendrig LM, Bossuyt PM, Dekker E. Participant-related risk factors for false-positive and false-negative fecal immunochemical tests in colorectal cancer screening: systematic review and meta-analysis. Am J Gastroenterol 2018;113(12):1778–87.
[53] Ungerer V, Bronkhorst AJ, Holdenrieder S. Preanalytical variables that affect the outcome of cell-free DNA measurements. Crit Rev Clin Lab Sci 2020;57(7):484–507.
[54] Orntoft MW, Jensen SO, Ogaard N, Henriksen TV, Ferm L, Christensen IJ, et al. Age-stratified reference intervals unlock the clinical potential of circulating cell-free DNA as a biomarker of poor outcome for healthy individuals and patients with colorectal cancer. Int J Cancer 2021;148(7):1665–75.
[55] Kotoh Y, Suehiro Y, Saeki I, Hoshida T, Maeda M, Iwamoto T, et al. Novel liquid biopsy test based on a sensitive methylated SEPT9 assay for diagnosing hepatocellular carcinoma. Hepatol Commun 2020;4(3):461–70.
[56] Lee HS, Hwang SM, Kim TS, Kim DW, Park DJ, Kang SB, et al. Circulating methylated septin 9 nucleic acid in the plasma of patients with gastrointestinal cancer in the stomach and Colon. Transl Oncol 2013;6(3):290–6.
[57] Schrock A, Leisse A, de Vos L, Gevensleben H, Droge F, Franzen A, et al. Free-circulating methylated DNA in blood for diagnosis, staging, prognosis, and monitoring of head and neck squamous cell carcinoma patients: an observational prospective cohort study. Clin Chem 2017;63(7):1288–96.
[58] de Klaver W, van der Vlugt M, Spaander MCW, Bossuyt PM, Dekker E. Risk of cancers proximal to the Colon in fecal immunochemical test positive screenees in a colorectal cancer screening program. Gastroenterology 2024;167(4):788–797 e2.
[59] Grobbee EJ, Schreuders EH, Hansen BE, Bruno MJ, Lansdorp-Vogelaar I, Spaander MCW, et al. Association between concentrations of hemoglobin determined by fecal immunochemical tests and long-term development of advanced colorectal neoplasia. Gastroenterology 2017;153(5):1251–9 e2.
[60] Green BB, BlueSpruce J, Tuzzio L, Vernon SW, Aubree Shay L, Catz SL. Reasons for never and intermittent completion of colorectal cancer screening after receiving multiple rounds of mailed fecal tests. BMC Public Health 2017;17(1):531.

[61] Ong C, Cook AR, Tan KK, Wang Y. Advancing colorectal cancer detection with blood-based tests: Qualitative study and discrete choice experiment to elicit population preferences. JMIR Public Health Surveill 2024;10:e53200.
[62] Schneider JL, Johnson CA, Jenkins C, Mummadi R, Coronado GD. "I was screaming hallelujah": patient and provider perceptions of blood-based testing for colorectal cancer screening. PLoS One 2023;18(12):e0295685.
[63] Zajac IT, Duncan A, Turnbull D, Wilson C, Flight I. Blood-based screening for bowel cancer may not resolve suboptimal screening participation in Australia. Aust N Z J Publ Health 2016;40(4):337–41.
[64] Benning TM, Dellaert BG, Dirksen CD, Severens JL. Preferences for potential innovations in non-invasive colorectal cancer screening: a labeled discrete choice experiment for a Dutch screening campaign. Acta Oncol 2014;53(7):898–908.
[65] Osborne J, Wilson C, Moore V, Gregory T, Flight I, Young G. Sample preference for colorectal cancer screening tests: blood or stool? OJPM, vol. 2; 2012. p. 326–31.
[66] Heidenreich S, Finney Rutten LJ, Miller-Wilson LA, Jimenez-Moreno C, Chua GN, Fisher DA. Colorectal cancer screening preferences among physicians and individuals at average risk: a discrete choice experiment. Cancer Med 2022;11(16):3156–67.
[67] Nayaradou M, Berchi C, Dejardin O, Launoy G. Eliciting population preferences for mass colorectal cancer screening organization. Med Decis Mak 2010;30(2):224–33.
[68] Moss S, Ancelle-Park R, Brenner H, International Agency for Research on C. European guidelines for quality assurance in colorectal cancer screening and diagnosis. First Edition–Evaluation and interpretation of screening outcomes. Endoscopy 2012;44(Suppl 3):SE49–64.
[69] Ding H, Lin J, Xu Z, Chen X, Wang HHX, Huang L, et al. A global evaluation of the performance indicators of colorectal cancer screening with fecal immunochemical tests and colonoscopy: a systematic review and meta-analysis. Cancers (Basel) 2022;14(4).
[70] Zhou Y, Li N, Luo J, Chen Y, Zhang Y, Dai M, et al. Participation and yield in multiple rounds of colorectal cancer screening based on fecal immunochemical test: a systematic review and meta-analysis. Am J Gastroenterol 2025;120(3):524–30.
[71] Le Bonniec A, Meade O, Fredrix M, Morrissey E, O'Carroll RE, Murphy PJ, et al. Exploring non-participation in colorectal cancer screening: a systematic review of qualitative studies. Soc Sci Med 2023;329:116022.
[72] Belon AP, McKenzie E, Teare G, Nykiforuk CIJ, Nieuwendyk L, Kim MO, et al. Effective strategies for Fecal Immunochemical Tests (FIT) programs to improve colorectal cancer screening uptake among populations with limited access to the healthcare system: a rapid review. BMC Health Serv Res 2024;24(1):128.
[73] Perisetti A, Khan H, George NE, Yendala R, Rafiq A, Blakely S, et al. Colorectal cancer screening use among insured adults: is out-of-pocket cost a barrier to routine screening? World J Gastrointest Pharmacol Therapeut 2018;9(4):31–8.
[74] Ahadinezhad B, Maleki A, Akhondi A, Kazemi M, Yousefy S, Rezaei F, et al. Are behavioral economics interventions effective in increasing colorectal cancer screening uptake: a systematic review of evidence and meta-analysis? PLoS One 2024;19(2):e0290424.
[75] Symonds EL, Cock C, Meng R, Cole SR, Fraser RJ, Young GP. Uptake of a colorectal cancer screening blood test in people with elevated risk for cancer who cannot or will not complete a faecal occult blood test. Eur J Cancer Prev 2018;27(5):425–32.
[76] Cole SR, Young GP, Byrne D, Guy JR, Morcom J. Participation in screening for colorectal cancer based on a faecal occult blood test is improved by endorsement by the primary care practitioner. J Med Screen 2002;9(4):147–52.
[77] Adler A, Geiger S, Keil A, Bias H, Schatz P, deVos T, et al. Improving compliance to colorectal cancer screening using blood and stool based tests in patients refusing screening colonoscopy in Germany. BMC Gastroenterol 2014;14:183.
[78] Coronado GD, Jenkins CL, Shuster E, Johnson C, Amy D, Cook J, et al. Blood-based colorectal cancer screening in an integrated health system: a randomised trial of patient adherence. Gut 2024;73(4):622–8.
[79] Liang PS, Zaman A, Kaminsky A, Cui Y, Castillo G, Tenner CT, et al. Blood test increases colorectal cancer screening in persons who declined colonoscopy and fecal immunochemical test: a randomized controlled trial. Clin Gastroenterol Hepatol 2023;21(11). 2951-7.e2.
[80] Symonds EL, Hughes D, Flight I, Woodman R, Chen G, Ratcliffe J, et al. A randomized controlled trial testing provision of fecal and blood test options on participation for colorectal cancer screening. Cancer Prev Res 2019;12(9):631–40.
[81] von Karsa L, Patnick J, Segnan N, Atkin W, Halloran S, Lansdorp-Vogelaar I, et al. European guidelines for quality assurance in colorectal cancer screening and diagnosis: overview and introduction to the full supplement publication. Endoscopy 2013;45(1):51–9.
[82] Wu W, Huang J, Yang Y, Gu K, Luu HN, Tan S, et al. Adherence to colonoscopy in cascade screening of colorectal cancer: a systematic review and meta-analysis. J Gastroenterol Hepatol 2022;37(4):620–31.
[83] Ioannou S, Sutherland K, Sussman DA, Deshpande AR. Increasing uptake of colon cancer screening in a medically underserved population with the addition of blood-based testing. BMC Cancer 2021;21(1):966.
[84] Laven-Law G, Symonds EL, Winter JM, Chen G, Flight IH, Hughes-Barton D, et al. Comparing a fecal immunochemical test and circulating tumor DNA blood test for colorectal cancer screening adherence. J Gastroenterol Hepatol 2024;39(7):1267–76.
[85] Barre S, Leleu H, Benamouzig R, Saurin JC, Vimont A, Taleb S, et al. Cost-effectiveness analysis of alternative colon cancer screening strategies in the context of the French national screening program. Therap Adv Gastroenterol 2020;13:1756284820953364.
[86] Benamouzig R, Barré S, Saurin J-C, Leleu H, Vimont A, Taleb S, et al. Cost-effectiveness analysis of alternative colorectal cancer screening strategies in high-risk individuals. Therapeutic Advances in Gastroenterology 2021;14:17562848211002359.
[87] Ladabaum U, Mannalithara A, Weng Y, Schoen RE, Dominitz JA, Desai M, et al. Comparative effectiveness and cost-effectiveness of colorectal cancer screening with blood-based biomarkers (Liquid biopsy) vs fecal tests or colonoscopy. Gastroenterology 2024;167(2):378–91.
[88] Aziz Z, Wagner S, Agyekum A, Pumpalova YS, Prest M, Lim F, et al. Cost-Effectiveness of liquid biopsy for colorectal cancer screening in patients who are unscreened. JAMA Netw Open 2023;6(11):e2343392. e.
[89] Nascimento de Lima P, Matrajt L, Coronado G, Escaron AL, Rutter CM. Cost-effectiveness of noninvasive colorectal cancer screening in community clinics. JAMA Netw Open 2025;8(1):e2454938. e.
[90] van den Puttelaar R, Nascimento de Lima P, Knudsen AB, Rutter CM, Kuntz KM, de Jonge L, et al. Effectiveness and cost-effectiveness of colorectal cancer screening with a blood test that meets the centers for medicare & medicaid services coverage decision. Gastroenterology 2024;167(2):368–77.
[91] Ladabaum U, Alvarez-Osorio L, Rösch T, Brueggenjuergen B. Cost-effectiveness of colorectal cancer screening in Germany: current endoscopic and fecal testing strategies versus plasma methylated septin 9 DNA. Endosc Int Open 2014;2(2):E96–104.
[92] Peterse EFP, Meester RGS, de Jonge L, Omidvari A-H, Alarid-Escudero F, Knudsen AB, et al. Comparing the cost-effectiveness of innovative colorectal cancer screening tests. J Natl Cancer Inst: J Natl Cancer Inst 2021;113(2):154–61.
[93] Ladabaum U, Mannalithara A, Schoen RE, Dominitz JA, Lieberman D. Projected impact and cost-effectiveness of novel molecular blood-based or stool-based screening tests for colorectal cancer. Ann Intern Med 2024;177(12):1610–20.
[94] Lew JB, St John DJB, Macrae FA, Emery JD, Ee HC, Jenkins MA, et al. Evaluation of the benefits, harms and cost-effectiveness of potential alternatives to iFOBT testing for colorectal cancer screening in Australia. Int J Cancer 2018;143(2):269–82.
[95] Nascimento de Lima P, van den Puttelaar R, Knudsen AB, Hahn AI, Kuntz KM, Ozik J, et al. Characteristics of a cost-effective blood test for colorectal cancer screening. J Natl Cancer Inst: J Natl Cancer Inst 2024;116(10):1612–20.
[96] Lansdorp-Vogelaar I, Goede SL, Bosch LJW, Melotte V, Carvalho B, van Engeland M, et al. Cost-effectiveness of high-performance biomarker tests vs fecal immunochemical test for noninvasive colorectal cancer screening. Clin Gastroenterol Hepatol 2018;16(4). 504-12.e11.
[97] Parikh RB, Prasad V. Blood-based screening for Colon cancer: a disruptive innovation or simply a disruption? JAMA 2016;315(23):2519–20.
[98] Cancer Council Australia Colorectal Cancer Screening Working Party. Clinical practice guidelines for the prevention, early detection and management of colorectal cancer: risk and screening based on family history. Sydney: Cancer Council Australia; 2023.
[99] Davidson KW, Barry MJ, Mangione CM, Cabana M, Caughey AB, Davis EM, et al. Screening for colorectal cancer: US preventive services task force recommendation statement. JAMA 2021;325(19):1965–77.
[100] Ola I, Cardoso R, Hoffmeister M, Brenner H. Utilization of colorectal cancer screening tests across European countries: a cross-sectional analysis of the European health interview survey 2018-2020. Lancet Reg Health Eur 2024;41:100920.
[101] Dix M, Cohen-Woods S, Wassie MM, Winter JM, Wilson CJ, Young GP, et al. The impact of age on preferences for colorectal cancer surveillance strategies: are fecal immunochemical tests FIT for surveillance? Cancer Med 2025;14(5):e70723.